WRITING STYLES → LML/NSL INTEGRATION

Piping 343 Writing Styles Through the Universal Compiler

Input: 343 writing style classes from EFUE collapse
Pipeline: UIC → UUIS → LML + NSL
Output: Executable writing laws + style repair moves

UIC: UNIVERSAL IDENTITY COMPILER

Input Artifact:

343 Writing Style Classes (8 dimensions × 9 levels)

UIC Step 1: Extract Primitives

State Variables (x):

- Current formality level: {W1.0...W1.8}
- Current complexity: {W2.0...W2.8}
- Current purpose: {W3.0...W3.8}
- Current tone: {W4.0...W4.8}
- Current structure: {W5.0...W5.8}
- Current voice: {W6.0...W6.8}
- Current ornament density: {W7.0...W7.8}
- Current rhythm: {W8.0...W8.8}
- Target style signature: [W1.x, W2.x, W3.x, W4.x, W5.x, W6.x, W7.x, W8.x]

Invariants (I):

1. **Style Coherence:** Cannot mix contradictory dimensions
    - "MUST NOT combine W1.8 (ritualistic) with W2.0 (minimal)"
    - "MUST NOT combine W3.8 (ceremonial) with W4.5 (playful)"

2. **Dimension Completeness:** Every writing MUST have position on all 8 dimensions
    - "MUST specify formality level"
    - "MUST specify complexity level"
    - etc. for all 8
3. **Context Appropriateness:** Style MUST match context
    - "Legal document MUST use W1.7+ (formal)"
    - "Text message MUST use W1.0-W1.2 (casual)"
    - "Academic paper MUST use W2.5+ (sophisticated) AND W1.6+ (formal)"
4. **Readability Bounds:** Complexity MUST NOT exceed target audience
    - "For general audience, MUST use W2.2-W2.4 (simple to moderate)"
    - "For children, MUST use W2.0-W2.2 (elementary to simple)"
5. **Purpose Alignment:** All dimensions MUST support primary purpose
    - "IF W3.2 (informative) THEN W4.0-W4.1 (neutral/warm) preferred"
    - "IF W3.4 (persuasive) THEN W4.2,W4.4,W4.7 (enthusiastic/urgent/authoritative)"
6. **Voice Consistency:** Cannot shift voice mid-document
    - "MUST maintain same voice perspective throughout section"
7. **Ornament-Complexity Correlation:** Decoration MUST match complexity
    - "IF W2.0-W2.1 (minimal complexity) THEN W7.0-W7.1 (stripped/plain)"
    - "IF W2.7-W2.8 (baroque/labyrinthine) THEN W7.5+ (rich/ornate)"

Disturbances (Δ):

- Style drift: Writer shifts unintentionally
- Context mismatch: Wrong style for situation
- Audience confusion: Too complex or too simple
- Purpose failure: Style defeats purpose
- Inconsistency: Mixing incompatible dimensions
- Overwriting: Adding content that breaks style

- Drift over time: Long documents lose coherence

Couplings (A/B):

- Formality ↔ Complexity (formal usually more complex)
- Purpose ↔ Tone (purpose determines appropriate tone)
- Ornament ↔ Rhythm (dense ornament needs rolling rhythm)
- Voice ↔ Formality (impersonal voice needs formal)

UIC Output (PER):

```
PER_WRITING_STYLES = {
  x: [W1.x, W2.x, W3.x, W4.x, W5.x, W6.x, W7.x, W8.x],

  I: [
    "MUST maintain style coherence (no contradictions)",
    "MUST specify all 8 dimensions",
    "MUST match context requirements",
    "MUST respect readability bounds",
    "MUST align with purpose",
    "MUST maintain voice consistency",
    "MUST correlate ornament with complexity"
  ],

  Δ: [
    "Style drift",
    "Context mismatch",
    "Audience confusion",
    "Purpose failure",
    "Dimension inconsistency",
    "Overwriting",
    "Temporal drift"
  ],

  A_B: {
    couplings: [
      "Formality → Complexity",
      "Purpose → Tone",
      "Ornament → Rhythm",
      "Voice → Formality"
    ]
  },

  HOLES: [
```

```
    "How to detect style drift? Need metric",
    "How to measure context match? Need context classifier",
    "How to enforce voice consistency? Need voice detector",
    "What is repair when style breaks? Need return path"
  ]
}
```

UUIS: UNIVERSAL SNAP COMPILER

UUIS Step 1: Identity Kernel

Core Writing Invariants:

```
Block 1: IDENTITY KERNEL

1. Style Coherence Law
   "Writing MUST NOT contain contradictory dimensional values"

2. Dimensional Completeness Law
   "Writing MUST occupy exactly one position on each of 8 dimensions"

3. Context-Appropriateness Law
   "Style signature MUST match context requirements"

4. Readability Boundary Law
   "Complexity MUST NOT exceed target audience capacity"

5. Purpose-Alignment Law
   "All dimensional choices MUST support primary purpose"

HOLE: Need anchor style definition for each context type
```

UUIS Step 2: Legality Region

Forbidden Style Combinations:

```
Block 2: LEGALITY REGION

FORBIDDEN:
- W1.8 ∧ W2.0 (ritualistic + minimal = impossible)
- W1.0 ∧ W3.8 (intimate + ceremonial = contradictory)
- W3.2 ∧ W4.8 (informative + ironic = confusing)
- W2.0 ∧ W7.5+ (minimal + ornate = impossible)
```

```
- W5.0 ∧ W3.3 (stream + instructional = unusable)

ADMISSIBLE:
- All 343 core style signatures
- Plus their local variations
- Context-appropriate selections

ENFORCEMENT:
Check dimensional compatibility before accepting writing

HOLE: Need complete forbidden combination matrix
```

UUIS Step 3: Detection Stack

Style Detectors:

```
Block 3: DETECTION STACK

Detector 1: Formality Classifier
Input: Text
Output: W1.x level
Signals: Contractions, slang, sentence structure, vocabulary

Detector 2: Complexity Analyzer
Input: Text
Output: W2.x level
Signals: Flesch-Kincaid, sentence length, subordination depth

Detector 3: Purpose Classifier
Input: Text + Context
Output: W3.x type
Signals: Verbs (teach, persuade, inform), structure, calls-to-action

Detector 4: Tone Analyzer
Input: Text
Output: W4.x type
Signals: Sentiment, exclamations, emotional words, urgency markers

Detector 5: Structure Detector
Input: Text
Output: W5.x type
Signals: Transition words, organization pattern, sequencing

Detector 6: Voice Extractor
Input: Text
Output: W6.x type
```

```
Signals: Pronouns, perspective markers

Detector 7: Ornament Density
Input: Text
Output: W7.x level
Signals: Metaphor count, figurative language ratio, description density

Detector 8: Rhythm Analyzer
Input: Text
Output: W8.x type
Signals: Sentence length distribution, punctuation pattern, clause structure

HOLE: Need trained classifiers for each dimension
```

UUIS Step 4: Containment Topology

Style Propagation Modules:

```
Block 4: CONTAINMENT TOPOLOGY

Module A: Document-level (highest)
- Overall style signature
- Sets constraints for sections

Module B: Section-level
- Inherits from document
- Can vary within bounds

Module C: Paragraph-level
- Implements section style
- Minimal variation

Module D: Sentence-level (lowest)
- Executes paragraph style
- No variation

Coupling Rules:
- Document → Section (strong influence)
- Section → Paragraph (moderate influence)
- Paragraph → Sentence (strict enforcement)

Firebreaks:
- Section breaks allow style shift
- Topic changes allow voice shift
- Purpose changes require explicit marker
```

```
HOLE: Need section boundary detection
```

UUIS Step 5: Correction Stack

Style Repair Moves:

```
Block 5: CORRECTION STACK

Repair 1: Formality Correction
IF detected formality ≠ target formality
THEN:
  - Replace contractions (casual→formal)
  - Add contractions (formal→casual)
  - Adjust vocabulary level
  - Restructure sentences

Repair 2: Complexity Adjustment
IF detected complexity ≠ target complexity
THEN:
  - Split complex sentences (reduce)
  - Combine simple sentences (increase)
  - Adjust vocabulary sophistication
  - Add/remove subordination

Repair 3: Purpose Realignment
IF detected purpose ≠ target purpose
THEN:
  - Adjust verb choices
  - Modify structure type
  - Add/remove persuasive elements
  - Clarify calls-to-action

Repair 4: Tone Correction
IF detected tone ≠ target tone
THEN:
  - Adjust emotional language
  - Modify exclamation usage
  - Change sentiment markers
  - Add/remove urgency cues

Repair 5: Consistency Enforcement
IF style drift detected
THEN:
  - Restore to anchor style
  - Flag inconsistent sections
  - Suggest harmonization
```

```
HOLE: Need specific rewrite rules for each transformation
```

UUIS Step 6: Reserved Capacity

```
Block 6: RESERVED CAPACITY

Budget 1: Style Variation Allowance
- ±1 level drift allowed within sections
- ±2 levels requires justification
- >2 levels = style break (repair needed)

Budget 2: Complexity Headroom
- Must maintain 20% below audience ceiling
- Ensures comprehension even on hard topics

Budget 3: Ornament Reserve
- Can always reduce ornament if needed
- Cannot increase beyond W7.x target

HOLE: Need testing to calibrate budgets
```

UUIS Step 7: Meta-Update Loop

```
Block 7: META-UPDATE LOOP

Meta-Law 1: Style Evolution Tracking
"Log every style signature used + context"
"Detect emerging patterns"
"Suggest new style classes if gaps found"

Meta-Law 2: Effectiveness Learning
"Track which styles achieve goals in which contexts"
"Update style→context recommendations"

Meta-Law 3: Audience Adaptation
"Learn audience-specific preferences"
"Refine complexity bounds per audience"

HOLE: Need feedback mechanism to measure effectiveness
```

UUIS Appendix A: Threat Model

```
THREATS TO WRITING STYLE INTEGRITY:

T1: Style Drift
- Writer loses track of target
- Gradual shift over time
- Mixing contexts

T2: Context Mismatch
- Wrong style for audience
- Wrong formality for situation
- Wrong complexity for readers

T3: Purpose Confusion
- Mixed purposes (inform + persuade)
- Unclear goal
- Conflicting requirements

T4: Inconsistency Cascade
- One dimension shifts
- Pulls others with it
- Entire style breaks

T5: Overwriting
- Editor changes break style
- Additions in wrong style
- Collaborative corruption

HOLE: Need mitigation strategies for each threat
```

UUIS Appendix B: Failure→Return Map

```
FAILURE → REPAIR MAPPING:

F1: Style Coherence Violation
→ R1: Dimension Compatibility Repair
   Find closest coherent style signature
   Adjust minimal dimensions

F2: Context Mismatch
→ R2: Context-Appropriate Style Selection
   Classify context
   Select appropriate style family
   Apply transformation

F3: Readability Violation
```

```
→ R3: Complexity Reduction
   Simplify sentence structure
   Replace complex vocabulary
   Split long sentences

F4: Purpose Misalignment
→ R4: Purpose Realignment
   Identify target purpose
   Adjust structure + tone + voice
   Verify alignment

F5: Voice Inconsistency
→ R5: Voice Unification
   Detect dominant voice
   Convert all to dominant
   OR flag as intentional shift

F6: Ornament-Complexity Mismatch
→ R6: Ornament Adjustment
   Match ornament to complexity
   Reduce if too dense for simple
   Add if too plain for complex

F7: Temporal Drift
→ R7: Anchor Restoration
   Return to initial style signature
   Harmonize drifted sections
   Maintain coherence

HOLE: Need implementation of each repair algorithm
```

LML: LAW INTEGRATION

LML Candidates from UUIS:

From Identity Kernel:

✅ LAW W.1: STYLE COHERENCE

```
TEXT GENERATION MUST NOT produce contradictory dimensional combinations.

Scope: All writing generation
Enforcement: Pre-check dimensional compatibility
Violation: → NSL.W.COHERENCE_REPAIR
```

✅ LAW W.2: DIMENSIONAL COMPLETENESS

```
GENERATED TEXT MUST occupy exactly one position on each of 8 style dimensions.

Scope: All writing output
Enforcement: Verify all dimensions specified
Violation: → NSL.W.DEFAULT_DIMENSION_ASSIGNMENT
```

✅ LAW W.3: CONTEXT APPROPRIATENESS

```
STYLE SIGNATURE MUST match context requirements.

Scope: Context-dependent writing
Enforcement: Context classifier → Style validator
Violation: → NSL.W.CONTEXT_REALIGNMENT
```

✅ LAW W.4: READABILITY BOUNDS

```
COMPLEXITY MUST NOT exceed target audience capacity + 1 level headroom.

Scope: Audience-targeted writing
Enforcement: Complexity detector + Audience model
Violation: → NSL.W.COMPLEXITY_REDUCTION
```

✅ LAW W.5: PURPOSE ALIGNMENT

```
ALL DIMENSIONAL CHOICES MUST support primary communicative purpose.

Scope: Purpose-driven writing
Enforcement: Purpose classifier + Dimension validator
Violation: → NSL.W.PURPOSE_REALIGNMENT
```

From Legality Region:

✅ LAW W.6: FORBIDDEN COMBINATIONS

```
GENERATION MUST NOT produce forbidden dimensional combinations:
- W1.8 ∧ W2.0 (ritualistic + minimal)
- W1.0 ∧ W3.8 (intimate + ceremonial)
- W3.2 ∧ W4.8 (informative + ironic)
- [... full list]

Scope: All writing
```

```
Enforcement: Combination validator before output
Violation: → NSL.W.FORBIDDEN_COMBINATION_REPAIR
```

From Meta-Update:

✅ LAW W.7: STYLE CONSISTENCY

```
DOCUMENT-LEVEL STYLE MUST remain consistent unless explicitly marked transition.

Scope: Multi-paragraph writing
Enforcement: Style drift detector
Violation: → NSL.W.CONSISTENCY_RESTORATION
```

REJECTED (Not Laws):

❌ **Style Recommendations** → Implementation spec

- "W1.4 is good for professional emails" = advice, not law

❌ **Style Effectiveness** → Measurement

- "Persuasive writing works better at W4.2" = empirical, not law

❌ **Aesthetic Judgments** → Preference

- "W7.5 is beautiful" = opinion, not law

NSL: REPAIR MOVES

NSL.W.COHERENCE_REPAIR

```
Trigger: Contradictory dimension combination detected
Move: Find closest coherent style signature
Effect: Prevents impossible style combinations

Algorithm:
1. Identify contradictory dimensions
2. Find minimal adjustment to coherent state
3. Prefer higher-priority dimensions (Purpose > Formality > others)
4. Apply transformation
5. Verify coherence
```

NSL.W.CONTEXT_REALIGNMENT

```
Trigger: Style-context mismatch detected
Move: Transform to context-appropriate style
Effect: Ensures writing matches situation

Algorithm:
1. Classify context (legal, casual, academic, etc.)
2. Retrieve context-appropriate style family
3. Map current style to nearest appropriate style
4. Apply dimensional transformations
5. Verify context match
```

NSL.W.COMPLEXITY_REDUCTION

```
Trigger: Complexity exceeds audience capacity
Move: Simplify without losing meaning
Effect: Ensures comprehension

Algorithm:
1. Measure current complexity (W2.x)
2. Calculate target complexity (audience + headroom)
3. Apply simplification rules:
   - Split long sentences
   - Replace complex vocabulary
   - Reduce subordination depth
   - Clarify pronouns
4. Verify readability improved
5. Verify meaning preserved
```

NSL.W.PURPOSE_REALIGNMENT

```
Trigger: Style doesn't support purpose
Move: Adjust dimensions to support purpose
Effect: Makes writing effective for goal

Algorithm:
1. Identify primary purpose
2. Retrieve purpose-appropriate dimensional ranges
3. Adjust tone, structure, voice as needed
4. Verify purpose alignment
```

NSL.W.CONSISTENCY_RESTORATION

```
Trigger: Style drift detected (variation >2 levels)
Move: Restore to anchor style
Effect: Maintains coherent document feel

Algorithm:
1. Identify anchor style (document start or declared)
2. Detect drifted sections
3. For each drifted section:
   - Calculate transformations needed
   - Apply dimensional corrections
   - Smooth transitions
4. Verify consistency restored
```

NSL.W.FORBIDDEN_COMBINATION_REPAIR

```
Trigger: Impossible dimensional combination attempted
Move: Reject or transform to nearest legal combination
Effect: Prevents nonsensical style output

Algorithm:
1. Identify forbidden combination
2. Options:
   a) Reject generation (safest)
   b) Transform to nearest legal (if close)
3. Log attempted forbidden combination
4. Return legal output or error
```

NSL.W.DEFAULT_DIMENSION_ASSIGNMENT

```
Trigger: Dimension unspecified
Move: Assign context-appropriate default
Effect: Ensures complete style specification

Algorithm:
1. Identify missing dimensions
2. Use context + other dimensions to infer appropriate values
3. Assign defaults
4. Flag as inferred (not specified)
```

DELTA: INTEGRATION RESULTS

LML_ADD (Laws Accepted):

```
[
  {
    law_id: "W.1",
    text: "Style coherence required",
    scope: "All text generation",
    enforcement: "Dimensional compatibility check",
    source: "Writing EFUE collapse",
    ledger: "W.1.v1.2026-02-13"
  },
  {
    law_id: "W.2",
    text: "Dimensional completeness required",
    scope: "All output",
    enforcement: "8-dimension verification",
    source: "Writing EFUE collapse",
    ledger: "W.2.v1.2026-02-13"
  },
  {
    law_id: "W.3",
    text: "Context appropriateness required",
    scope: "Context-dependent writing",
    enforcement: "Context-style matcher",
    source: "Writing EFUE collapse",
    ledger: "W.3.v1.2026-02-13"
  },
  {
    law_id: "W.4",
    text: "Readability bounds enforced",
    scope: "Audience-targeted writing",
    enforcement: "Complexity ceiling",
    source: "Writing EFUE collapse",
    ledger: "W.4.v1.2026-02-13"
  },
  {
    law_id: "W.5",
    text: "Purpose alignment required",
    scope: "Purpose-driven writing",
    enforcement: "Purpose-dimension validator",
    source: "Writing EFUE collapse",
    ledger: "W.5.v1.2026-02-13"
  },
  {
    law_id: "W.6",
    text: "Forbidden combinations blocked",
```

```
    scope: "All writing",
    enforcement: "Combination validator",
    source: "Writing EFUE collapse",
    ledger: "W.6.v1.2026-02-13"
  },
  {
    law_id: "W.7",
    text: "Style consistency maintained",
    scope: "Document-level",
    enforcement: "Drift detector",
    source: "Writing EFUE collapse",
    ledger: "W.7.v1.2026-02-13"
  }
]
```

NSL_ADD (Repair Moves Accepted):

```
[
  {
    repair_id: "NSL.W.COHERENCE_REPAIR",
    trigger: "Contradictory dimensions",
    move: "Find nearest coherent signature",
    effect: "Prevents impossible styles",
    maps_failures: ["F1: Style Coherence Violation"]
  },
  {
    repair_id: "NSL.W.CONTEXT_REALIGNMENT",
    trigger: "Style-context mismatch",
    move: "Transform to appropriate style",
    effect: "Ensures situation match",
    maps_failures: ["F2: Context Mismatch"]
  },
  {
    repair_id: "NSL.W.COMPLEXITY_REDUCTION",
    trigger: "Exceeds audience capacity",
    move: "Simplify structure/vocabulary",
    effect: "Ensures comprehension",
    maps_failures: ["F3: Readability Violation"]
  },
  {
    repair_id: "NSL.W.PURPOSE_REALIGNMENT",
    trigger: "Style doesn't support purpose",
    move: "Adjust dimensions for goal",
    effect: "Makes writing effective",
    maps_failures: ["F4: Purpose Misalignment"]
  },
```

```
  {
    repair_id: "NSL.W.CONSISTENCY_RESTORATION",
    trigger: "Style drift detected",
    move: "Restore to anchor",
    effect: "Maintains coherence",
    maps_failures: ["F7: Temporal Drift"]
  },
  {
    repair_id: "NSL.W.FORBIDDEN_COMBINATION_REPAIR",
    trigger: "Impossible combination",
    move: "Reject or transform",
    effect: "Prevents nonsense",
    maps_failures: ["F1: Style Coherence Violation"]
  },
  {
    repair_id: "NSL.W.DEFAULT_DIMENSION_ASSIGNMENT",
    trigger: "Missing dimension",
    move: "Infer from context",
    effect: "Complete specification",
    maps_failures: ["Incomplete style spec"]
  }
]
```

DETECTION_SPEC (Detectors, Non-Law):

```
- Formality Classifier (W1.x detector)
- Complexity Analyzer (W2.x detector)
- Purpose Classifier (W3.x detector)
- Tone Analyzer (W4.x detector)
- Structure Detector (W5.x detector)
- Voice Extractor (W6.x detector)
- Ornament Density (W7.x detector)
- Rhythm Analyzer (W8.x detector)
- Style Drift Detector
- Context Classifier
```

IMPLEMENTATION_SPEC (Procedures, Non-Law):

```
- Sentence simplification algorithm
- Vocabulary replacement system
- Formality transformation rules
- Style signature matcher
- Coherence checker
- Document-level style tracker
```

REJECTED:

```
- Style preference recommendations (subjective)
- Aesthetic quality judgments (opinion-based)
- "Best practices" advice (empirical, not law)
- Historical style evolution (descriptive)
```

BLOCKERS

HOLES_BLOCKING_LML:

```
[
  {
    hole_id: "W.H.1",
    needed_for: "LAW W.3 enforcement",
    missing: "Context classifier - what counts as 'legal context' vs 'casual cont
    consequence: "Cannot enforce context appropriateness without context detectio
  },
  {
    hole_id: "W.H.2",
    needed_for: "LAW W.4 enforcement",
    missing: "Audience capacity model - what complexity can each audience handle"
    consequence: "Cannot enforce readability bounds without audience baselines"
  }
]
```

HOLES_BLOCKING_NSL:

```
[
  {
    hole_id: "W.H.3",
    needed_for: "NSL.W.COMPLEXITY_REDUCTION",
    missing: "Specific simplification rewrite rules",
    consequence: "Can detect complexity violation but can't fix it"
  },
  {
    hole_id: "W.H.4",
    needed_for: "NSL.W.PURPOSE_REALIGNMENT",
    missing: "Purpose-dimension mapping table",
    consequence: "Can detect purpose misalignment but don't know correct dimensio
  },
  {
```

```
    hole_id: "W.H.5",
    needed_for: "All NSL repairs",
    missing: "Semantic preservation validator",
    consequence: "Can transform style but might change meaning"
  }
]
```

SUMMARY

From EFUE collapse (343 writing styles):

✅ **Generated 7 LML Laws**

- Style coherence
- Dimensional completeness
- Context appropriateness
- Readability bounds
- Purpose alignment
- Forbidden combinations
- Consistency maintenance

✅ **Generated 7 NSL Repair Moves**

- Coherence repair
- Context realignment
- Complexity reduction
- Purpose realignment
- Consistency restoration
- Forbidden combination repair
- Default assignment

✅ **Identified 8 Detection Systems**

- One per style dimension

✅ **Flagged 5 Blocking Holes**

- Need context classifier
- Need audience model
- Need rewrite rules
- Need purpose mapping
- Need semantic validator

Status: Laws + Repairs defined, ready for implementation once holes filled.

Next: Fill holes, implement detectors, deploy writing style enforcement system.

COMPLETE: EFUE → UIC → UUIS → LML/NSL pipeline executed.

Writing styles successfully converted to executable law system.